Motor TPL mock rate filing (freMTPL2 public data)

Contents

  Exhibit 1 Overall rate indication and assumptions
  Exhibit 2 Current vs proposed relativities, credibility and caps
  Exhibit 3 Premium dislocation
  Exhibit 4 Model support

Every assumption in Exhibit 1 except experience losses and exposure is illustrative.
The holdout is a random 80/20 split stratified by claim count; the data carry no policy dates,
so this is not a forward in time test.

Policies: 668,874; claims: 26,142; exposure years: 353721.
Overall indicated change: +5.8%.
Severity relativities used: FALSE (Gamma GLM vs flat mean on holdout: −0.03% (95% CI −0.87% to 0.85%)).
Policies whose premium moves more than 10% either way (revenue neutral structure): 55.7%.

Exhibit 1. Overall rate indication (loss ratio method)

assumption	value	basis
Experience losses	60280571	estimated: capped losses x excess factor
Exposure	353721	data
Large loss cap	35651 per claim	estimated: 99.5th pct of train claims
Excess loss factor	1.3601	estimated: train
Loss development factor	1.00	assumed (data treated as ultimate)
Annual loss trend	3% for 2 years	illustrative
ULAE load	10% of loss	illustrative
Variable expense ratio	25%	illustrative
Profit and contingency	5%	illustrative
Current premium	94943651	assumed: incumbent plan priced at experience period cost, no trend

Projected loss and LAE ratio = losses x LDF x (1 + trend)^years x (1 + ULAE) / premium = 0.7409
Permissible loss ratio = 1 − 0.25 − 0.05 = 0.70
Indicated overall change = 0.7409 / 0.70 − 1 = +5.85%

The current premium is the incumbent plan priced at experience period cost, so the indication is driven by the trend assumption plus any gap between incumbent expected and actual losses.

Exhibit 2. Current vs proposed relativities

indicated = GLM frequency relativity x GLM severity relativity (severity = 1 when the severity GLM fails holdout, as here);
Z = min(1, sqrt(claims / 1082));
credibility weighted = Z x indicated + (1 − Z) x current; final = credibility weighted, change capped at +/− 15%.

var	level	exposure	claims	indicated	current	Z	cred	final	change_pct	capped
VehPowerB	6	65267	4973	1.000	1.000	1.00	1.000	1.000	0.0	FALSE
VehPowerB	4	47363	3149	0.931	1.000	1.00	0.931	0.931	−6.9	FALSE
VehPowerB	5	53714	4079	0.983	1.000	1.00	0.983	0.983	−1.7	FALSE
VehPowerB	7	61297	4392	0.997	1.000	1.00	0.997	0.997	−0.3	FALSE
VehPowerB	8	18046	1315	1.034	1.000	1.00	1.034	1.034	3.4	FALSE
VehPowerB	9	37175	2997	1.189	1.000	1.00	1.189	1.150	15.0	TRUE
VehAgeB	6−10	78165	6314	1.000	1.000	1.00	1.000	1.000	0.0	FALSE
VehAgeB	0	12499	878	0.983	1.000	0.90	0.985	0.985	−1.5	FALSE
VehAgeB	1−2	49535	3658	1.043	1.000	1.00	1.043	1.043	4.3	FALSE
VehAgeB	3−5	56966	4250	1.008	1.000	1.00	1.008	1.008	0.8	FALSE
VehAgeB	11−15	61636	4465	0.877	1.000	1.00	0.877	0.877	−12.3	FALSE
VehAgeB	16+	24060	1340	0.705	1.000	1.00	0.705	0.850	−15.0	TRUE
DrivAgeB	41−50	69525	5323	1.000	1.000	1.00	1.000	1.000	0.0	FALSE
DrivAgeB	18−20	2031	482	1.137	1.108	0.67	1.127	1.127	1.7	FALSE
DrivAgeB	21−25	10810	1406	0.704	0.705	1.00	0.704	0.704	−0.1	FALSE
DrivAgeB	26−30	23106	1865	0.557	0.564	1.00	0.557	0.557	−1.4	FALSE
DrivAgeB	31−40	66520	4638	0.692	0.699	1.00	0.692	0.692	−0.9	FALSE
DrivAgeB	51−60	59737	4095	0.952	0.956	1.00	0.952	0.952	−0.4	FALSE
DrivAgeB	61−70	30259	1857	0.900	0.887	1.00	0.900	0.900	1.6	FALSE
DrivAgeB	71+	20873	1239	0.937	0.893	1.00	0.937	0.937	5.0	FALSE
BonusMalusB	50	177259	9161	1.000	1.000	1.00	1.000	1.000	0.0	FALSE
BonusMalusB	51−60	38031	2841	1.608	1.634	1.00	1.608	1.608	−1.6	FALSE
BonusMalusB	61−70	20986	2490	2.592	2.654	1.00	2.592	2.592	−2.3	FALSE
BonusMalusB	71−80	21031	2174	2.394	2.431	1.00	2.394	2.394	−1.6	FALSE
BonusMalusB	81−90	12128	1368	2.610	2.621	1.00	2.610	2.610	−0.4	FALSE
BonusMalusB	91−100	10577	1880	3.961	3.950	1.00	3.961	3.961	0.3	FALSE
BonusMalusB	101+	2849	991	7.775	7.865	0.96	7.779	7.779	−1.1	FALSE
VehBrand	B1	75299	5468	1.000	1.000	1.00	1.000	1.000	0.0	FALSE
VehBrand	B10	7539	594	0.977	1.000	0.74	0.983	0.983	−1.7	FALSE
VehBrand	B11	5456	530	1.200	1.000	0.70	1.140	1.140	14.0	FALSE
VehBrand	B12	50750	3235	0.742	1.000	1.00	0.742	0.850	−15.0	TRUE
VehBrand	B13	5423	445	1.047	1.000	0.64	1.030	1.030	3.0	FALSE
VehBrand	B14	1789	108	0.850	1.000	0.32	0.953	0.953	−4.7	FALSE
VehBrand	B2	74753	5405	0.993	1.000	1.00	0.993	0.993	−0.7	FALSE
VehBrand	B3	22695	1929	1.038	1.000	1.00	1.038	1.038	3.8	FALSE
VehBrand	B4	10878	874	1.007	1.000	0.90	1.006	1.006	0.6	FALSE
VehBrand	B5	15759	1310	1.063	1.000	1.00	1.063	1.063	6.3	FALSE
VehBrand	B6	12523	1007	1.029	1.000	0.96	1.028	1.028	2.8	FALSE
VehGas	Regular	147831	10221	1.000	1.000	1.00	1.000	1.000	0.0	FALSE
VehGas	Diesel	135031	10684	1.171	1.000	1.00	1.171	1.150	15.0	TRUE
Area	C	82225	5604	1.000	1.000	1.00	1.000	1.000	0.0	FALSE
Area	A	48842	2645	0.845	1.000	1.00	0.845	0.850	−15.0	TRUE
Area	B	33927	2087	0.929	1.000	1.00	0.929	0.929	−7.1	FALSE
Area	D	60913	5120	1.150	1.000	1.00	1.150	1.150	15.0	TRUE
Area	E	50515	4847	1.232	1.000	1.00	1.232	1.150	15.0	TRUE
Area	F	6441	602	1.263	1.000	0.75	1.196	1.150	15.0	TRUE
Region	R24	80678	5152	1.000	1.000	1.00	1.000	1.000	0.0	FALSE
Region	R11	23818	2051	1.005	1.138	1.00	1.005	1.005	−11.6	FALSE
Region	R21	946	51	0.845	0.792	0.22	0.803	0.803	1.5	FALSE
Region	R22	2816	235	1.073	1.130	0.47	1.103	1.103	−2.3	FALSE
Region	R23	2465	175	0.986	1.048	0.40	1.023	1.023	−2.4	FALSE
Region	R25	5257	351	0.962	1.023	0.57	0.988	0.988	−3.4	FALSE
Region	R26	3948	265	0.991	0.968	0.49	0.979	0.979	1.2	FALSE
Region	R31	9092	753	1.022	1.140	0.83	1.041	1.041	−8.6	FALSE
Region	R41	6454	376	0.911	0.988	0.59	0.943	0.943	−4.6	FALSE
Region	R42	953	64	0.908	1.019	0.24	0.992	0.992	−2.6	FALSE
Region	R43	452	29	0.888	0.863	0.16	0.867	0.867	0.5	FALSE
Region	R52	17418	1252	1.017	1.089	1.00	1.017	1.017	−6.5	FALSE
Region	R53	21745	1447	0.980	1.055	1.00	0.980	0.980	−7.0	FALSE
Region	R54	8870	658	1.085	1.106	0.78	1.089	1.089	−1.5	FALSE
Region	R72	11367	835	1.053	1.103	0.88	1.059	1.059	−4.0	FALSE
Region	R73	5517	292	0.783	0.790	0.52	0.786	0.786	−0.4	FALSE
Region	R74	1907	146	1.285	1.202	0.37	1.233	1.233	2.5	FALSE
Region	R82	36004	3375	1.203	1.344	1.00	1.203	1.203	−10.5	FALSE
Region	R83	1832	105	0.885	0.848	0.31	0.860	0.860	1.4	FALSE
Region	R91	11581	810	0.980	1.005	0.87	0.984	0.984	−2.1	FALSE
Region	R93	28363	2381	1.112	1.233	1.00	1.112	1.112	−9.8	FALSE
Region	R94	1377	102	1.109	1.026	0.31	1.052	1.052	2.5	FALSE

Exhibit 3. Premium dislocation

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 band         share_of_policies
 below −10%   0.2901
 −10% to +10% 0.4430
 above +10%   0.2668

Quantiles of proposed / current − 1 (revenue neutral structure):
1% −31.8%   10% −19.9%   50% −0.6%   90% +21.8%   99% +41.1%
The overall +5.8% change applies on top of these.
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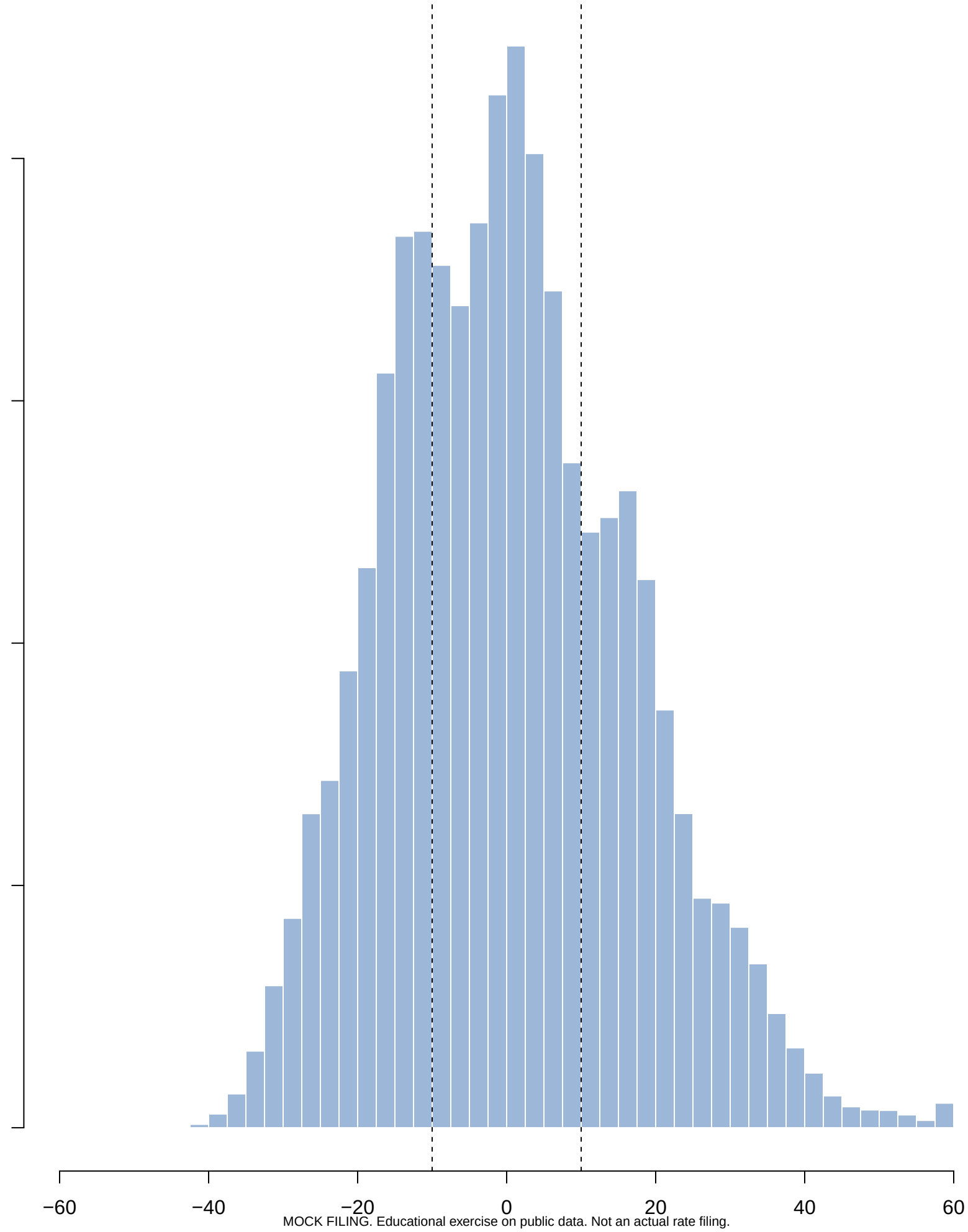
Premium change, revenue neutral structure



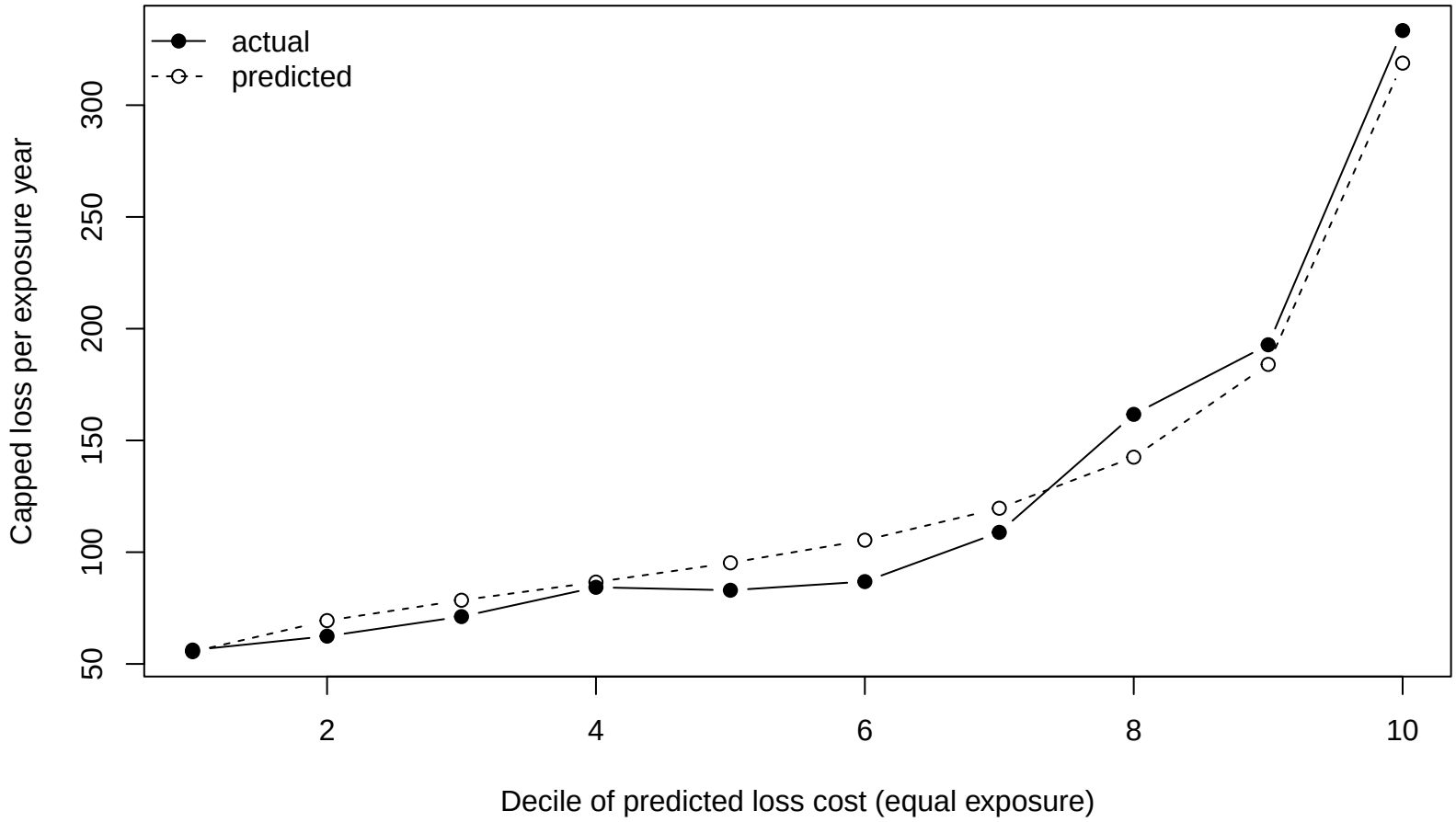
MOCK FILING. Educational exercise on public data. Not an actual rate filing.

Exhibit 4a. Holdout model comparison (test set)

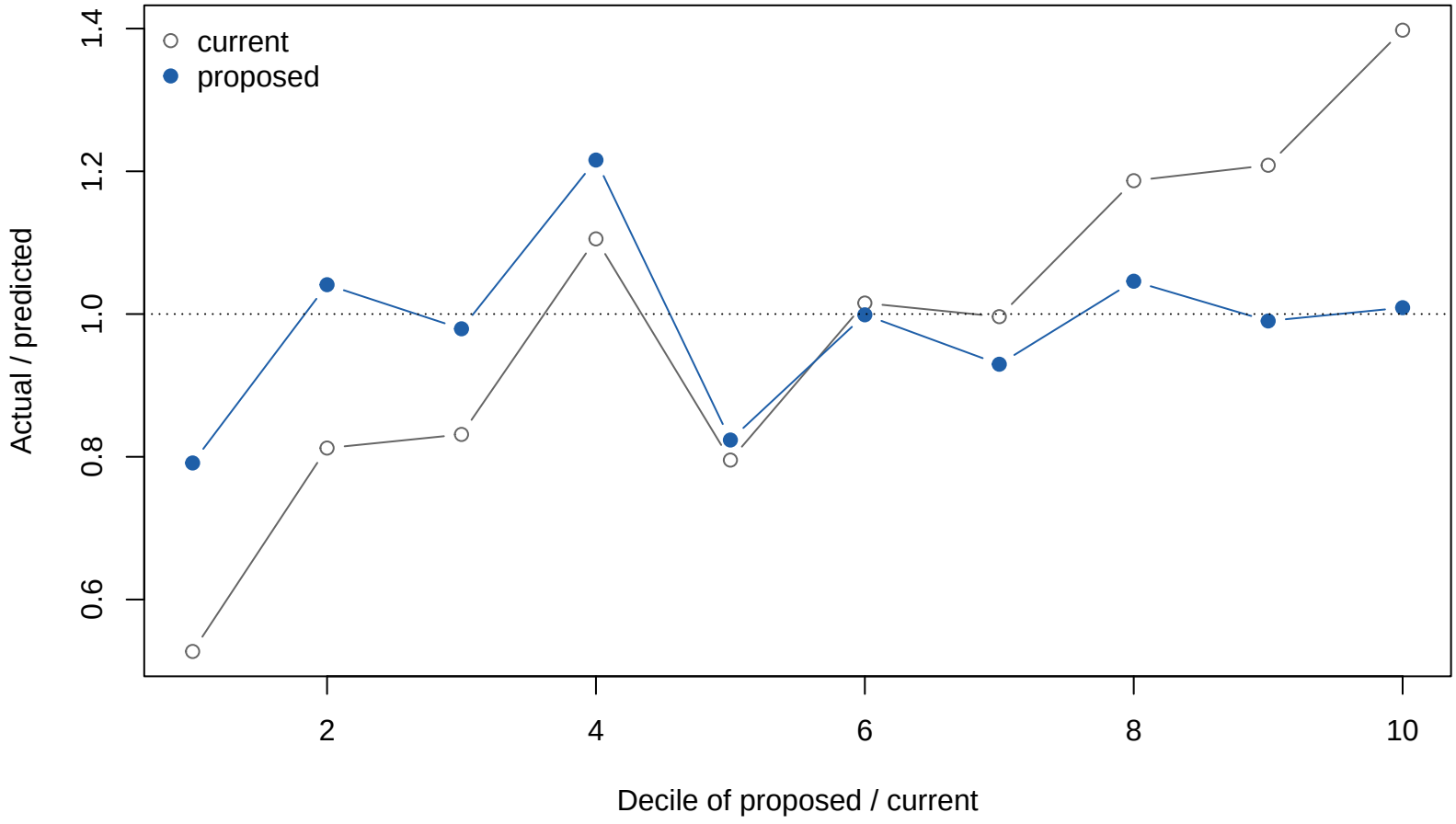
comparison	pct_deviance_reduction	lo95	hi95
proposed_glm_vs_exposure_only	5.086	4.560	5.596
proposed_glm_vs_current_glm	0.545	0.354	0.726
gbm_vs_proposed_glm	1.684	1.377	2.003
glm_interaction_vs_proposed	0.354	0.238	0.482
gbm_vs_glm_interaction	1.335	1.028	1.651
severity_glm_vs_mean	−0.032	−0.867	0.850

model	gini
current GLM (frequency)	0.2809
proposed GLM (frequency)	0.3024
GBM (frequency)	0.3347
current plan (pure premium)	0.2942
proposed GLM (pure premium)	0.3264

Lift, proposed plan, test set



Double lift, current vs proposed pure premium



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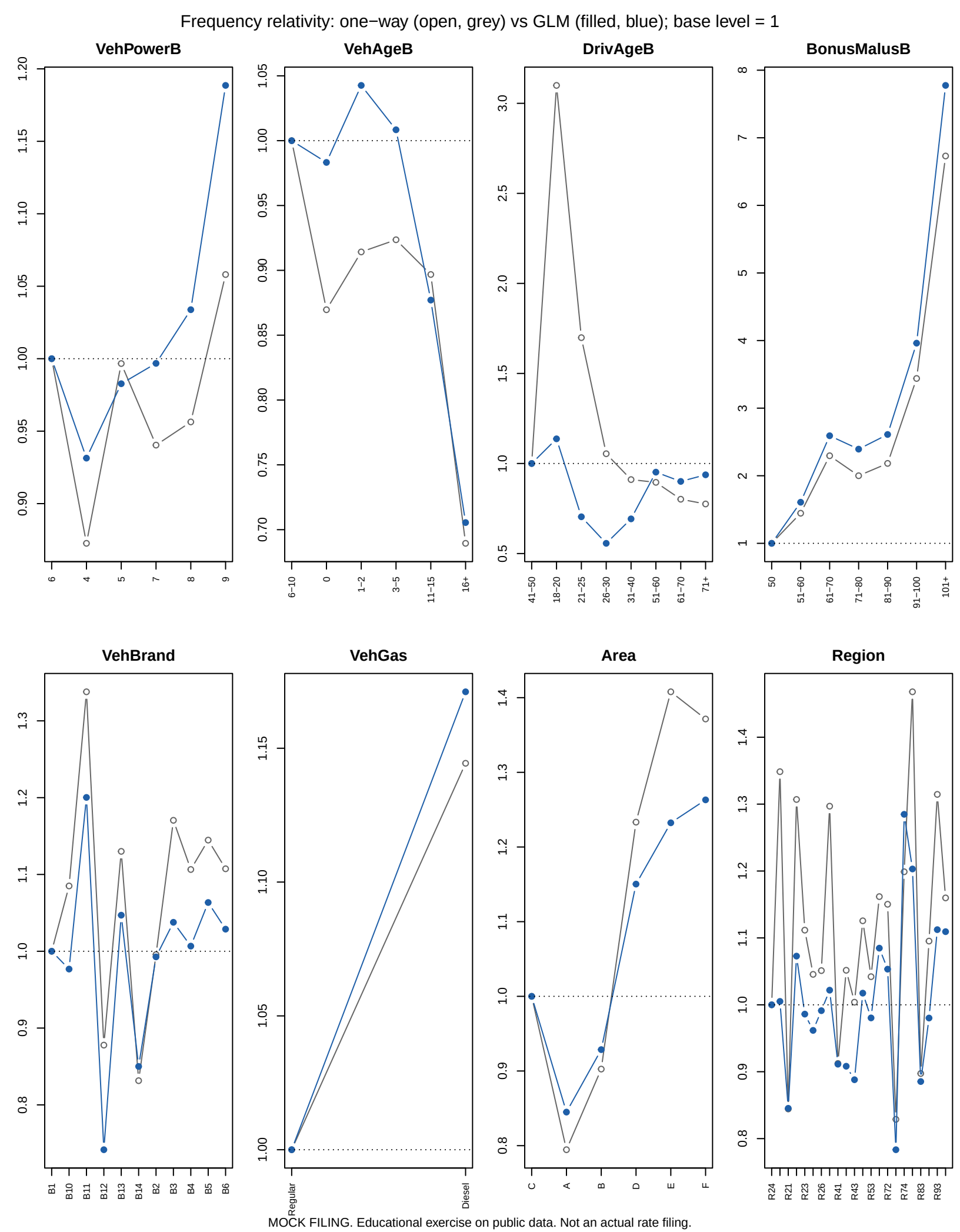

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Exhibit 4b. Proposed frequency GLM coefficients (Poisson, log link, exposure offset)

term	estimate	se	relativity
(Intercept)	−2.9386	0.0312	0.0529
VehPowerB4	−0.0711	0.0241	0.9314
VehPowerB5	−0.0174	0.0216	0.9827
VehPowerB7	−0.0032	0.0213	0.9968
VehPowerB8	0.0333	0.0318	1.0338
VehPowerB9	0.1728	0.0251	1.1886
VehAgeB0	−0.0169	0.0375	0.9832
VehAgeB1−2	0.0417	0.0224	1.0426
VehAgeB3−5	0.0084	0.0202	1.0084
VehAgeB11−15	−0.1312	0.0198	0.8771
VehAgeB16+	−0.3489	0.0306	0.7055
DrivAgeB18−20	0.1284	0.0515	1.1370
DrivAgeB21−25	−0.3508	0.0334	0.7041
DrivAgeB26−30	−0.5857	0.0287	0.5567
DrivAgeB31−40	−0.3675	0.0207	0.6925
DrivAgeB51−60	−0.0492	0.0209	0.9520
DrivAgeB61−70	−0.1049	0.0271	0.9005
DrivAgeB71+	−0.0649	0.0320	0.9372
BonusMalusB51−60	0.4749	0.0225	1.6079
BonusMalusB61−70	0.9523	0.0237	2.5918
BonusMalusB71−80	0.8728	0.0259	2.3935
BonusMalusB81−90	0.9595	0.0314	2.6104
BonusMalusB91−100	1.3766	0.0294	3.9614
BonusMalusB101+	2.0509	0.0360	7.7747
VehBrandB10	−0.0235	0.0450	0.9768
VehBrandB11	0.1826	0.0467	1.2003
VehBrandB12	−0.2989	0.0257	0.7416
VehBrandB13	0.0459	0.0498	1.0470
VehBrandB14	−0.1625	0.0975	0.8501
VehBrandB2	−0.0073	0.0193	0.9927
VehBrandB3	0.0370	0.0268	1.0377
VehBrandB4	0.0067	0.0368	1.0067
VehBrandB5	0.0615	0.0312	1.0635
VehBrandB6	0.0284	0.0346	1.0288
VehGasDiesel	0.1579	0.0150	1.1711
AreaA	−0.1685	0.0245	0.8450
AreaB	−0.0740	0.0258	0.9286
AreaD	0.1400	0.0197	1.1503
AreaE	0.2090	0.0211	1.2324
AreaF	0.2336	0.0513	1.2631
RegionR11	0.0051	0.0330	1.0051
RegionR21	−0.1681	0.1411	0.8453
RegionR22	0.0703	0.0673	1.0728
RegionR23	−0.0140	0.0772	0.9861
RegionR25	−0.0390	0.0554	0.9618
RegionR26	−0.0088	0.0632	0.9913
RegionR31	0.0217	0.0403	1.0219
RegionR41	−0.0929	0.0543	0.9113
RegionR42	−0.0963	0.1263	0.9082
RegionR43	−0.1187	0.1864	0.8880
RegionR52	0.0173	0.0320	1.0175
RegionR53	−0.0198	0.0303	0.9804
RegionR54	0.0813	0.0416	1.0847
RegionR72	0.0519	0.0379	1.0533
RegionR73	−0.2441	0.0608	0.7834
RegionR74	0.2506	0.0841	1.2848
RegionR82	0.1849	0.0237	1.2031
RegionR83	−0.1218	0.0990	0.8853
RegionR91	−0.0199	0.0386	0.9803
RegionR93	0.1065	0.0268	1.1124
RegionR94	0.1038	0.1009	1.1093

Exhibit 4c. Severity GLM coefficients (Gamma, log link, capped average claim)

term	estimate	se	relativity
(Intercept)	7.3466	0.0387	1550.9835
DrivAgeB18−20	0.3101	0.1000	1.3636
DrivAgeB21−25	−0.0142	0.0655	0.9859
DrivAgeB26−30	−0.0052	0.0557	0.9948
DrivAgeB31−40	0.0071	0.0397	1.0071
DrivAgeB51−60	0.0387	0.0399	1.0395
DrivAgeB61−70	0.0221	0.0518	1.0223
DrivAgeB71+	0.2100	0.0608	1.2336
VehAgeB0	0.1443	0.0693	1.1553
VehAgeB1−2	0.0775	0.0401	1.0805
VehAgeB3−5	0.0171	0.0382	1.0173
VehAgeB11−15	−0.0344	0.0377	0.9661
VehAgeB16+	−0.0512	0.0581	0.9501
BonusMalusB51−60	0.0510	0.0425	1.0523
BonusMalusB61−70	−0.0313	0.0444	0.9692
BonusMalusB71−80	0.1207	0.0490	1.1283
BonusMalusB81−90	0.0911	0.0596	1.0953
BonusMalusB91−100	0.2002	0.0584	1.2216
BonusMalusB101+	0.1091	0.0697	1.1153
VehGasDiesel	−0.0018	0.0270	0.9982